

COURSE TITLE : WOOD PRODUCTS-II
COURSE CODE : 5122
COURSE CATEGORY : A
PERIODS PER WEEK : 4
SEMESTER : 5
PERIODS PER SEMESTER : 52
CREDITS : 4

TIME SCHEDULE

Module	Topics	Periods
I	Block Board	4
	Flush door	4
	Properties and Testing	4
	Test	1
II	Particle Board	6
	Drying of particle Board	5
	Test	1
III	Storage of Chips	5
	Chip Blending	4
	Mat Formation	1
	Test	1
IV	Pressing of particle Board	5
	Finishing	4
	Calender particle Board	4
	Properties and testing	2
	Test	1
	Total	52

Course Outcome :

Module	G.O	Student will be able to:
1	1	Understand the Manufacture of Block Board
	2	Understand the Method of Manufacture of Flushdoor
	3	Know the Testing of Flushdoor and Blockboard
2	1	Understand Particleboard
	2	Understand Drying of Particle Board
3	1	Understand Chip Storage and Chip Blending
4	1	Understand Pressing and Finishing of Particle Board
	2	Know Properties and Testing

Specific outcome

MODULE-I

1.1.0 Understand the Manufacture of Block Board

- 1.1.1 State lamin board, block board, battern board
- 1.1.2 Describe core, cross bands and faces used in block boards
- 1.1.3 Describe manufacturing process of block board by strip method and block method
- 1.1.4 Specify the assembling of block board core
- 1.1.5 Explain continuous gluing and spot gluing
- 1.1.6 Explain decorative, non decorative and block board

1.2.0 Understand the Method of Manufacture of Flushdoor

- 1.2.1 State flush door and their function
- 1.2.2 Describe solid core, solid particle board core, hollow core doors
- 1.2.3 Explain adhesives used in flush door manufacturing
- 1.2.4 Explain lipping and rebate cutting

1.3.0 Know the Testing of Flushdoor and Blockboard

- 1.3.1 Estimate alternative wetting and drying test knife test, boiling test of flush door
- 1.3.2 Estimate alternative wetting and drying test, knife test and boiling test of block board as per BIS

MODULE II

2.1.0 Understand Particle Board

- 2.1.1 Define particles, flakes, shavings, splinter and sliver, granules, state history of production of P.B
- 2.1.2 Describe exterior grade, interior grade, single layer, three layer, multi layer, graded, extrusion pressed and flat presses P.B
- 2.1.3 List the raw materials used for manufacturing of P.B
- 2.1.4 Describe about the UF and PF resin used in P.B manufacture and state general rules of chip resination
- 2.1.5 Describe the additives used in P.B manufacture
- 2.1.6 Explain the schematic outline for P.B manufacture
- 2.1.7 Define debarking and list and describe the names of debarking machines
- 2.1.8 List the machines used for cross cutting and splitting of wood.
- 2.1.9 Explain the working of cutter spindle chipper disc type chipper and toothed disc Mills

2.2.0 Understand Drying of Particle Board

- 2.2.1 State the need of Drying of PB
- 2.2.2 Describe drying of chips

- 2.2.3 State the advantages of drying of chips
- 2.2.4 List the classification dryers
- 2.2.5 Describe rotary drum dryer & multiple band dryer, burned waste gas stream dryer and suspension type dryer

MODULE III

3.1.0 Understand Chip Storage and Chip Blending

- 3.1.1 State the need of Chip storage and Drying of Chip
- 3.1.2 Explain the various screening and sifting methods
- 3.1.3 Describe the horizontal, vertical and rotating storage bins
- 3.1.4 List the chip conveying methods
- 3.1.5 Define chip blending
- 3.1.6 State general considerations of chip chip blending
- 3.1.7 Describe batch operation and continuous operation one each
- 3.1.8 Describe chip spreading (mat laying)
- 3.1.9 State the terms and failures in volumetric and gravimetric dosing of the materials
- 3.1.10 Describe dosing of resonated chips by purely volumetric dosing and control of the Weight of the chips
- 3.1.11 Explain the working of spreading heads with necessary diagram
- 3.1.12 Describe the working of spreading machine with necessary diagram
- 3.1.13 Describe the resultant mat.

MODULE IV

4.1.0 Understand Pressing and Finishing of Particle Board

- 4.1.1 State Pressing of Particleboard
- 4.1.2 State the advantages and disadvantages of the compression roller pre pressing System
- 4.1.3 List the two wetting methods of resultant mat
- 4.1.4 State the four particular proceeding of the hot platen press
- 4.1.5 State the three methods applied in single daylight press to active short pressing time
- 4.1.6 Describe the effect of press platen temperature on physical properties of P.B
- 4.1.7 Describe the effect of curing time on physical properties of P.B
- 4.1.8 Describe the influence of quality of hardener and proportion of formaldehyde in the urea resin on quality of P.B
- 4.1.9 Distinguish the characteristics of phenolic resin with urea formaldehyde resin
- 4.1.10 Describe the continues belt particle board production system
- 4.1.11 Explain the chip craft – vertical extrusion process
- 4.1.12 Explain the Lane wood horizontal extrusion process
- 4.1.13 Describe Calendar particleboard Process

4.2.0 Know Properties and Testing

- 4.2.1 State the physical and chemical changes take place during the storage of boards after leaving the press
- 4.2.2 Describe board sizing
- 4.2.3 Explain sanding or thicknessing of P.B using wide belt sanding machine
- 4.2.4 Describe the preparation and conditioning of test specimen
- 4.2.5 Specify the factors affecting the properties of P.B
- 4.2.6 Specify the physical properties, general properties, mechanical properties, technological properties and resistance against destruction.
- 4.2.7 Describe the following testing methods of P.B, Density , Moisture content, absorption and swelling, Bending strength, Compression strength , perpendicular to the plane of board, modulus of elasticity, Tensile strength perpendicular to the plane of board.
- 4.2.8 State the test result of a particular item of P.B as per international standards

MODULE I

Block board Types - block method, strip method. Assembling of block board core. Continuous gluing, spot gluing. Decorative and non-decorative block board Flush door Types – solid core flush door, cellular core (cellulose) flush door. Adhesive used. Lipping and rebate cutting, facing Testing and properties of block board & flush door Properties of block board & flush door. Alternative wetting and drying test, knife test, boiling test of Blockboard

MODULE II

Particle board – introduction – history – particles, flakes, shavings, splinter and sliver, and granules. Classification of P.B – exterior grade, interior grade, single layer, three layer , multi layer, graded, flat pressed and extension pressed P.B uses of P.B, Raw materials , adhesive – UF and PF resin, additives – water repellents, extenders, fungicides and insecticides and fire retardants, Schematic out line P.B name factors. Particle preparation, debarking cross cutting, splitting Classification of dryers, rotary drum dryers and tube bundle dryers, burned waste gas stream dryer, suspension type dryer.

MODULE III

Storage of chips – Horizontal, vertical and rotating storage bins, screening and sifting methods, chip conveying methods. Chip blending – Definition, general consideration, batch operation, continuous operation, Mat laying – Terms and failures in volumetric and gravimetric dosing. Dosing of resinated chips. Volumetric and gravimetric dosing. Dosing of resinated chips volumetric dosing and control of the weight of the chips. Spreader heads and spreading machines, resultant mat

MODULE IV

Pre-pressing and pressing of P.B – pre pressing methods – advantages and its disadvantages. Pressing – two wetting methods of mat, four particular proceedings of the hot platen press. Three methods to

achieve short pressing time in single day light press- Effect of press platen temperature on physical properties of P.B. Effect of curing time on physical properties of P.B, influences of quality of hardner and proportion of formaldehyde in the urea resin on quality of P.B. characteristics of UF resin with PF resin. List 3 Caul less pressing process – continuous belt system. Chip craft vertical extrusion process, lane wood horizontal extrusion processes Calender Particleboard-Mande-Baisen System-P.B finishing – Sizing sanding or thicknessing by wide belt sanding machine. Physical and chemical changes during storage of P.B after leaving press. Properties and Testing of P.B – Preparation and conditioning of test specimen, factors affecting the properties of P.B. Physical properties, general properties, mechanical properties, technological properties and reistance against destruction. P.B testing methods – Density, moisture content, absorption and swelling, bending strength, compression shear strength perpendicular to the plane of board, Modulus of elasticity, tensile strength perpendicular to the plane of board. State standard test result of P.B

REFERENCE

- 1.Franz. F.P Kollmann ,Principles of wood science and technology Vol. II - Springer
- 2.Thomas M Malon , Modern particle board and drying process of fiber board manufacturing –